GENERAL THETA FOUNDATIONS I: ACQUIRED GEOMETRY AND CAUSAL RESOURCE TRANSFER

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Abstract. We prove an acquired-geometry transfer theorem for positive causal instruments with state-dependent likelihoods and finite executable test closure. Bayesian normalization replaces the transported quadratic energy by a convex perspective potential. The loss of predictive energy has an exact decomposition into its conditional Jensen defect, within-row quantization, and the additional defect of pooling incompatible predecessor centroids. This gives a converse for every Borel randomized finite-state observer. Actual conditional acquisition mass and a global covering certificate, together with either integrated-curvature stability or stable posterior transport, give matching checkpoint and online laws. With an internally enforced deadline of n acquisitions and M total states, the excess is comparable to $(M-n)^{-2/d}$, uniformly in both budgets; the checkpoint excess is comparable to $M^{-2/d}$. The dimension is verified from the acquired report map. Positive Gaussian hidden-state filters and informative noncommuting quantum instruments verify the same theorem, without Haar report laws. Periodically positivity-improving quantum instruments need not contract at each step. A full-rank/pure-state instrument mixture yields a rank-changing, actual-mass resource profile. We also prove a matched calibration–deficiency extension in which retaining an independently acquired bit changes the internal-clock charge, and give separate finite-precision and causal-simulator certificates.

1. The experiment, rather than a parameter manifold

A finite observer does not possess the posterior that appears in its analysis. It sees the current report and its retained label. Its next label must serve all histories that have been pooled into the old one. This distinction becomes decisive when a report has a state-dependent likelihood: the conditional law of the forgotten history changes on observing the report, and a linear transport formula is no longer valid.

The problem considered here starts with a preparation, positive causal report instruments, and a declared family of executable future tests. Their conditional evaluations form a predictive quotient. The squared task fixes a norm on these evaluations. No covariance, Fisher matrix, spectrum, or smooth parameter manifold is supplied as a geometric axiom. The relevant measures are the laws actually produced by the preparation and report protocol, including the conditional laws entering each finite-state update.

Our central result is Theorem 2.2. Its first part is an exact identity valid without a stability assumption. For a normalized posterior map $F(y,u) = U(y,u)/\ell(y,u)$, where U and ℓ are affine before normalization, introduce

$$\Phi(u) = \int \frac{\|U(y,u)\|^2}{\ell(y,u)}\,\lambda(dy).$$

The map is convex. Its Hessian, when differentiation is justified, is the report-law average of the squared derivative of the normalized prediction. If X is the full-history predictive state and $\bar{X}$ its retained conditional mean, the error propagated through a report is exactly the conditional

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Jensen defect of Φ. The remaining loss divides into quantization within each entering label and disagreement between the row centroids assigned to a common outgoing label.

This identity separates three questions that static quantization alone cannot answer: optimality within a row, compatibility of rows at one update, and existence of a compatible recursion through an entire experiment. The lower bound treats all Borel randomized observers. The upper bound is a genuine finite-state construction. It uses either calibrated common cells, with stability measured by the integrated likelihood curvature, or an explicitly stable posterior recursion on a global cover. Analytical conditional means do not become free runtime registers.

The second part of the theorem gives a sharp statistical resource law. The certificates are stated before choosing an observer: actual one-step acquisition small balls, a global cover, and one of two verifiable stability conditions. Under common positive report support, an internally timed observer must pay for every reachable phase. The resulting excess is

$$\mathcal{R}_{\mathrm{aut}}(n, M) - B_n \asymp (M - n)^{-2/d}, \qquad M \geq n + 1.$$

Here B_n is the full-history Bayes risk of the same prescribed experiment. This is not an assertion that arbitrary active sensing or optimal stopping has been solved. The kernels may vary with the prescribed phase, but no external phase is given to the autonomous observer. A checkpoint encoder has access to the complete history once and satisfies the different law $\mathcal{R}_{\mathrm{cp}}(n, M) - B_n \asymp M^{-2/d}$.

The two principal verifications are genuinely informative, non-Haar experiments. The classical model is a positive, invertible hidden-state transition observed through a nondegenerate Gaussian likelihood. Its conditional acquisition dimension is derived from the Gaussian log-odds map. The quantum model uses full-dimensional weak effects followed by a positivity-improving channel. Its report law depends on the entering density operator, and its effects do not share an eigenbasis. A positive congruence inverse verifies its acquisition mass; a cone metric verifies stable execution. A periodically improving version permits noncontracting stages. A second quantum instrument with a pure-state branch gives a genuine rank-changing acquired measure and retains each stratum's physical mass.

Finally, Theorem 8.3 shows why errors and memory cannot be combined by notation alone. An independently acquired bit, its actual report erasure, and an unobservable calibration offset are placed in the same terminal task. Remembering that bit through an internally timed experiment changes the available phase budget. The theorem proves matching upper and all-observer lower bounds, not merely an additive error estimate.

Conditional expectation, convex perspectives, projective contraction, and static quantization are standard tools. The contribution claimed here is their instrument-level causal combination, its all-observer row/pooling converse, and the jointly counted finite-state constructions. The relation to sequential vector quantization, finite-state coding, recursive marginal quantization, and information-rate converses is made precise in Section 9. The earlier relative-report energy and exact-circle results are retained in full as Companion G; the temporal theory remains in separate complete companions. None of those realizations is used to prove the present general theorem backwards.

2. Normalized experiments and the central theorem

2.1. **Raw tests and their affine closure.** Fix a known preparation and a prescribed finite report protocol of length $n \geq 1$. All physical state, action and report spaces are standard Borel. The visible report law is generated by normalized positive causal kernels; positive operator instruments are also allowed through their ordinary classical report laws. One preparation, n reports, and one environment-owned terminal audit are $N = n + 2$ paid physical calls. The answer is committed before the audit. That audit is not a report available to the observer.

A finite *executable affine closure certificate* consists of the constant test and k bounded test coordinates, a compact convex prediction domain $K_t \subset \mathbb{R}^k$ for each phase, a Borel realization of those coordinates, and, for λ_t-almost every report y, affine maps

$$\ell_t(y,u) = \ell_t^0(y) + c_t(y)u, \qquad U_t(y,u) = a_t(y) + A_t(y)u.$$

The raw joint report/test identities give, for every realized incoming prediction u,

$$\mathbb{P}(Y_t \in dy \mid u) = \ell_t(y,u)\lambda_t(dy), \qquad F_t(y,u) = \frac{U_t(y,u)}{\ell_t(y,u)} \in K_t. \tag{1}$$

In particular $\int \ell_t(y,u)\lambda_t(dy) = 1$. The reference measure may be sigma-finite. Likelihoods are positive on a common full-measure set for the main resource theorem. The identity theorem permits zero likelihood, using the closed convex perspective and setting its value to zero when both numerator and likelihood vanish. The identities hold on the whole declared domain, not only at the states of a proposed quantizer. They can be verified by pulling the raw instrument back on the finite tests. All matrix coordinates are obtained from those tests; they are not postulated state geometry.

Write $\mathcal{H}_t$ for the complete report history and X_t for its predictive vector. Then $X_t = F_t(Y_t, X_{t-1})$, with known X_0. A performed bounded vector audit Z_n satisfies $\mathbb{E}[Z_n \mid \mathcal{H}_n] = X_n$. Its Euclidean squared norm is the fixed task metric, with response set K_n; a larger closed convex response set containing K_n gives the same optimum. Set

$$\nu_t = \mathrm{Law}(X_t), \qquad B_n = \mathbb{E}\,\|Z_n - X_n\|^2. \tag{2}$$

These are laws and risks under the actual preparation and protocol, independent of the observer. There is no infimum over a different sensing protocol hidden in B_n.

Lemma 2.1 (Realized predictive quotient)**.** *Suppose the declared future tests contain the coordinate tests and their legal report continuations, and their evaluations are determined by the affine instrument certificate. If the evaluation image is standard Borel and has a Borel realization, equality of the coordinates and of the visible action/clock signature is exactly predictive equivalence for these tests. The quotient is measurable, realizable and minimal among Borel statistics sufficient for them. Jointly Borel action-indexed certificates descend with the legal action signature unchanged.*

Proof. Iterating the unnormalized instrument identities determines the probabilities of all finite report words and all declared terminal test expectations. A monotone-class argument extends from a countable generating family of report events to its sigma-field. Conversely the included coordinate tests distinguish different vectors. Any sufficient Borel statistic carries each coordinate as a Borel function, and hence carries their finite vector. Evaluation at the assumed Borel realization gives descended kernels, independently of the representative; composition preserves joint action measurability. The interface signature prevents identifying histories with different legal continuations. These assertions concern the declared test family, not unmodeled physical tasks. □

For the finite hidden-state and density-operator models below, mixtures realize the entire compact convex domain. The deterministic map sending a state coordinate to its physical preparation is a Borel realization. Restricting to a Borel attained image gives the same task quotient; a formal ambient set is never substituted for its acquired probability law.

2.2. **Finite observers and certificates.** A scheduled observer has finite alphabets S_t, $|S_t| \le m_t$, and Borel rows $k_t(j \mid s, y)$. Only S_{t-1}, Y_t and fresh independent randomness are available at the update. A persistent seed, history tape, current exact posterior or real-valued auxiliary register is not free. Put

$$V_t = \mathbb{E}[X_t \mid S_t], \qquad e_t = \mathbb{E}\,\|V_t\|^2, \qquad r_t = \mathbb{E}\,\|X_t - V_t\|^2.$$

The best response based on S_n is V_n, and total risk is $B_n + r_n$. Zero-mass labels are discarded in sums. A fixed Borel program may contain exact constants; its bit implementation is a separate resource statement in Section 8.

An autonomous observer has one finite label set, one reused report row, one initial law and a halt/readout map. It must itself halt exactly after n acquisitions almost surely. The physical experiment continues to make reports available until it halts. No external end marker or phase is supplied. All labels, including phases and halt labels, count toward M. The report spaces are then one common space with equivalent positive conditional laws at every possible phase and history. This is a support condition, not equality of report probabilities.

For $u \in K_{t-1}$ let

$$\mu_{t,u} = (F_t(\cdot, u))_{\#}(\ell_t(\cdot, u)\lambda_t), \qquad \Phi_t(u) = \int \|z\|^2 \, \mu_{t,u}(dz). \tag{3}$$

At an actual incoming label with $V_{t-1} = u$, $\mu_{t,u}$ is the law of the prediction after seeing the report but before replacing the label. A uniform certificate on K_{t-1} is a way of controlling every actual such row, not a claim that every point of K_{t-1} has been acquired.

Fix $d > 0$. The geometric certificates, with constants independent of n, t and the observer, are:

$$\mu_{t,u}(B(z,h)) \le C_{\mathrm{mass}} h^d \qquad (u \in K_{t-1},\ z \in \mathbb{R}^k,\ 0 < h \le h_0), \tag{4}$$

$$\mathcal{N}(K_t, \mathrm{dist}_t, h) \le C_{\mathrm{cov}} h^{-d}. \tag{5}$$

The cover is global and consists of legal representatives in K_t; by enlarging constants it supplies an at-most-m codebook of radius $Cm^{-1/d}$ for every integer $m \ge 1$. Distances and diameters are uniformly bounded and $\|u - v\| \le C \, \mathrm{dist}_t(u, v)$. In the curvature alternative below, dist_t is Euclidean. Report continuity is certified separately by

$$\|\ell_t(\cdot, u)\lambda_t - \ell_t(\cdot, v)\lambda_t\|_{\mathrm{TV}} \le L_{\mathrm{rep}} \|u - v\|. \tag{6}$$

It is needed for robust implementation and composition, not to replace the actual measure in (4).

One of the following *stability certificates* is required for the matching upper.

(C): Φ_t is twice differentiable on an open relative neighborhood of K_{t-1} (or by continuous extension from its relative interior), with

$$\tfrac{1}{2} D^2 \Phi_t(u)[v, v] \le b_t \|v\|^2, \qquad \prod_{j=s+1}^{t} b_j \le C_0 \rho^{t-s}, \quad 0 < \rho < 1.$$

The derivative interchange in (9) is justified by a declared integrable bound. Convexity supplies the lower sign; no positive Hessian lower is required.

(P): The executable normalized update satisfies

$$\mathrm{dist}_t(F_t(y, u), F_t(y, v)) \le a_t \, \mathrm{dist}_{t-1}(u, v), \qquad \prod_{j=s+1}^{t} a_j \le C_0 \rho^{t-s}$$

for common nonexceptional reports and all u, v. Individual a_t may equal or exceed one. A global Borel nearest-cell rule for (5) is part of the certificate.

Empty products equal one; $C_0 \ge 1$. These are experiment-level conditions. They do not assert that every positive experiment is stable.

Theorem 2.2 (Bayesian causal acquired-geometry transfer)**.** *For every prepared experiment satisfying the affine test certificate, every horizon $n \ge 1$, and every scheduled Borel randomized finite observer, the following exact identity holds at each $1 \le t \le n$:*

$$r_t = \mathbb{E}\Phi_t(X_{t-1}) - \mathbb{E}\Phi_t(V_{t-1}) + D_t^{\mathrm{row}} + D_t^{\mathrm{pool}}. \tag{7}$$

The last two terms are the nonnegative conditional-square losses defined in (13)*. The first term is a nonnegative conditional Jensen defect. In particular*

$$r_t \geq \sum_s \mathbb{P}(S_{t-1} = s) q_{m_t}(\mu_{t,V_{t-1}(s)}). \tag{8}$$

When differentiation is justified, the likelihood-induced curvature is

$$\tfrac{1}{2} D^2 \Phi_t(u)[v,v] = \int \ell_t(y,u) \, \|D_u F_t(y,u) v\|^2 \, \lambda_t(dy). \tag{9}$$

Assume, in addition, (4)–(6)*, the global domain/cover conditions, and (C) or (P). There are constants $0 < c \leq C < \infty$, depending only on the displayed certificate constants, such that for all $n \geq 1$, $m \geq 1$ and $M \geq n+1$,*

$$\mathcal{R}_{\mathrm{cp}}(n,m) = B_n + q_m(\nu_n), \qquad cm^{-2/d} \leq q_m(\nu_n) \leq Cm^{-2/d}, \tag{10}$$

$$c(M-n)^{-2/d} \leq \mathcal{R}_{\mathrm{aut}}(n,M) - B_n \leq C(M-n)^{-2/d}. \tag{11}$$

For (11) *require the autonomous common-support interface just specified. The autonomous class is empty for $M < n+1$ and nonempty for $M \geq n+1$. Its infimum is over all legal Borel randomized programs, not just the quantizers constructed in the proof. Both upper bounds are achieved up to the displayed constants by legal finite-state programs. In (P) the state carries a codepoint label; in (C) it carries a calibrated-cell label. Their programs and output precision are not included in the number of labels.*

The first part is proved in Section 3; the matching statements in Section 4. The theorem permits positive informative likelihoods, nonhomogeneous acquired laws and block, rather than per-step, stability. It does not claim an unrestricted optimal-control theorem: the preparation and physical report protocol are fixed before optimizing the observer. A controller reading additional history would change the experiment and requires a separate proof.

3. The Bayesian defect and causal compatibility

3.1. **The actual conditional experiment at an incoming label.** All conditional expectations below include the observer's previous independent randomizations. They do not include future audit outcomes. Fix a phase and suppress its index. For an incoming label s of positive probability write $u_s = \mathbb{E}[X \mid S = s]$. Affinity of ℓ and U gives, for every report event E,

$$\mathbb{P}(Y \in E \mid S = s) = \int_E \ell(y, u_s) \lambda(dy), \qquad \mathbb{E}[X' 1_{\{Y \in E\}} \mid S = s] = \int_E U(y, u_s) \lambda(dy).$$

Consequently, if $W = F(Y, u_S)$,

$$\mathbb{E}[X' \mid S, Y] = W. \tag{12}$$

This is a conditional Bayes identity, not an interchange of a nonlinear map and an expectation under the old history law. The current report reweights that law by its likelihood. If J is the new label, freshness of its update randomization implies $\mathbb{E}[X' \mid S, Y, J] = W$. The same formula follows first for bounded simple reports and then by conditional integration; it does not condition a continuous law on an event of positive singleton probability.

For positive-mass pairs (s, j) put

$$c_{sj} = \mathbb{E}[W \mid S = s, J = j], \qquad v_j = \mathbb{E}[W \mid J = j].$$

Define, with zero-mass summands omitted,

$$D^{\mathrm{row}} = \mathbb{E} \, \|W - c_{SJ}\|^2 , \qquad D^{\mathrm{pool}} = \mathbb{E} \, \|c_{SJ} - v_J\|^2 . \tag{13}$$

The tower property gives $v_J = \mathbb{E}[X' \mid J]$. Orthogonal conditional projections yield

$$\mathbb{E}\|W\|^2 = D^{\mathrm{row}} + D^{\mathrm{pool}} + \mathbb{E}\|v_J\|^2, \tag{14}$$

$$\mathbb{E}\|X'\|^2 = \mathbb{E}\Phi(X), \qquad \mathbb{E}\|W\|^2 = \mathbb{E}\Phi(u_S). \tag{15}$$

Subtracting proves (7). In particular the potential's Jensen term is exactly

$$\mathbb{E}\Phi(X) - \mathbb{E}\Phi(u_S) = \mathbb{E}\|X' - \mathbb{E}[X' \mid S, Y]\|^2. \tag{16}$$

This also proves its nonnegativity without differentiability. It measures the old information that remains inaccessible even after the new report has arrived.

3.2. **Perspective curvature.** For $q > 0$, the function $(w, q) \mapsto \|w\|^2 / q$ is convex, since

$$\frac{\|w\|^2}{q} = \sup_{z \in \mathbb{R}^k} \{2\langle z, w\rangle - q\|z\|^2\}.$$

The closed perspective at $(0, 0)$ has value zero. The bounded prediction domain implies $U = 0$ whenever $\ell = 0$, so this convention agrees with the raw experiment. Integration and affine pullback prove convexity of Φ on K. For positive likelihood and a tangent direction v, direct differentiation gives

$$\frac{1}{2}D^2\left(\frac{\|U\|^2}{\ell}\right)[v, v] = \frac{\|Av - F(cv)\|^2}{\ell} = \ell\|DF[v]\|^2. \tag{17}$$

The assumed domination justifies integration and proves (9). The formula is an acquired likelihood curvature: its weight is the physical report density, and its derivative is that of the normalized predictive update. It is not a separately specified Fisher metric.

When Φ is differentiable, define

$$D_\Phi(x, u) = \Phi(x) - \Phi(u) - D\Phi(u)[x - u].$$

Conditioning on S removes the last term and gives

$$\mathbb{E}\Phi(X) - \mathbb{E}\Phi(u_S) = \mathbb{E}D_\Phi(X, u_S). \tag{18}$$

If $\frac{1}{2}D^2\Phi \le bI$ on every segment in K, Taylor's integral formula, including its factor $\int_0^1 (1-s)ds = 1/2$, implies

$$0 \le \mathbb{E}D_\Phi(X, u_S) \le b\mathbb{E}\|X - u_S\|^2. \tag{19}$$

A measurable supporting affine function gives the analogous nonnegative identity on a nonsmooth domain whenever a subgradient with the required integrability is available. No such selection is needed for the unconditional identity (16).

3.3. **Converse and the three compatibility questions.** For a finite measure ζ on a compact Euclidean set write

$$q_m(\zeta) = \inf_{a_1, \dots, a_m} \int \min_j \|x - a_j\|^2 \, \zeta(dx).$$

Centers may lie in the closed convex hull; compactness makes the infimum attainable. For any randomized assignment to at most m labels and any assigned responses, conditional minimization gives distortion at least $q_m(\zeta)$. Conversely deterministic nearest-center assignment attains the center infimum. This argument applies to fractional cells as well as deterministic partitions.

At a fixed old label s, the law of W is μ_{t,u_s}, and the new label is a randomized assignment to at most m_t values. Therefore

$$D^{\mathrm{row}} \ge \sum_s \mathbb{P}(S = s) q_{m_t}(\mu_{t,u_s}).$$

Together with the nonnegative Jensen and pooling terms, this proves (8). The sharper statement retaining the Jensen term is immediate from the same equality.

For a *fixed incoming law and a fixed one-step labeling*, equality in this sharper bound holds if and only if every positive-mass row attains its static quantization optimum and all row centroids using the same outgoing label agree. Necessity follows by summing nonnegative losses; sufficiency follows from (14). This separates static row optimality and zero pooling. It does not prove that such labelings can be chosen compatibly at every phase. Even an attained static optimum in every row need not admit a common outgoing labeling. The common-cover construction in the next section instead bounds the *sum* of the two defects and proves a global matching rate without claiming local saturation.

In the state-independent linear case $\ell(y,u)=\ell(y)$, $F(y,u)=A(y)u$, and $\int A(y)^T A(y)\ell(y)\lambda(dy)=\kappa I$, one has $\Phi(u)=\kappa\|u\|^2$. Thus (7) reduces to

$$r_t=\kappa r_{t-1}+D^{\mathrm{row}}+D^{\mathrm{pool}},\qquad e_t=\kappa e_{t-1}-D^{\mathrm{row}}-D^{\mathrm{pool}}.$$

The exact energy and circle results of Companion G are recovered with their original assumptions. Zero-energy rows contribute zero; a later zero product factor does not certify earlier saturation. Their directional coefficient is an infimum, and equality requires an actual minimizing direction and row encoder, not merely approximate attainment.

Example 3.1 (An informative report cannot be treated as a neutral transport)**.** Prepare $X\in\{-1,1\}$ with equal probabilities and retain one old label. Let $0<\eta<1$ and report $Y\in\{-1,1\}$ with $\mathbb{P}(Y=X\mid X)=(1+\eta)/2$. A two-label update stores Y. Its retained posterior is ηY, so its retained energy rises from zero to η^2. For a prior mean u, the report likelihood is $(1+\eta yu)/2$ and the posterior mean is $(u+\eta y)/(1+\eta yu)$. Its potential is

$$\Phi(u)=\frac{1}{2}\sum_{y=\pm1}\frac{(u+\eta y)^2}{1+\eta yu}.$$

Here $\Phi(0)=\eta^2$, whereas $\Phi(1)=\Phi(-1)=1$. The identity gives residual error $1-\eta^2$ and zero row/pooling loss. It explains the increase in information through the likelihood term; a state-independent energy recursion would miss it.

4. From actual acquisition to a counted implementation

4.1. **Actual mass and checkpoint risk.** A probability ζ satisfying $\zeta(B(z,h))\le Ah^d$ obeys

$$q_m(\zeta)\ge c(A,d,h_0)m^{-2/d},\qquad m\ge1. \tag{20}$$

Indeed choose $h=c_0m^{-1/d}\le h_0$ so that $mAh^d\le1/2$. The union of m radius-h balls has mass at most $1/2$, so every codebook has distortion at least $h^2/2$. The same lower holds for randomized encoders by conditional minimization. This is a bound on the actual probability, not a volume argument on a merely reachable set.

The actual full-history law is $\nu_t=\int\mu_{t,u}\,\nu_{t-1}(du)$, so it inherits the same small-ball bound. A global cover of K_t gives the matching upper. Conditional projection onto the complete history and then onto a checkpoint label gives

$$\inf\mathbb{E}\,\|Z_n-a(S)\|^2=B_n+q_m(\nu_n).$$

Nearest-center coding of X_n attains the infimum over codebooks. A checkpoint is allowed to compute from the whole history; this is not yet an online program.

4.2. **Common-cover recursion.** Under (C), start at the known X_0. Suppose all positive incoming labels have programmed constants $u_s = \mathbb{E}[X_{t-1} \mid S_{t-1} = s]$. Choose a common at-most-m_t Euclidean codebook covering K_t to radius $h_t = Cm_t^{-1/d}$. From the actual old label and report compute $W = F_t(Y_t, u_s)$ and send it to its Borel nearest cell. Program the new representative as its conditional mean under this very observer's law. It lies in K_t by convexity. Finite forward integration determines these constants; the program does not compute a full-history posterior at runtime.

For this construction, regardless of how incoming rows overlap in an outgoing cell,

$$D_t^{\mathrm{row}} + D_t^{\mathrm{pool}} = \mathbb{E}\,\|W - \mathbb{E}[W \mid S_t]\|^2 \le \mathbb{E}\,\|W - a_{S_t}\|^2 \le h_t^2. \tag{21}$$

The competitor a_{S_t} is the common codepoint. Combining (19) and (21), and iterating from $r_0 = 0$, gives

$$r_n \le C \sum_{t=1}^{n} \left(\prod_{j=t+1}^{n} b_j \right) m_t^{-2/d}. \tag{22}$$

Exact Borel implementability permits exact calibrated constants; computability and bit precision are separately addressed below. No independent rowwise quantizer is assumed to have zero pooling loss.

Under (P), choose an at-most-m_t global codebook in the declared metric. The observer stores only $\widehat{X}_t = Q_t(F_t(Y_t, \widehat{X}_{t-1}))$. Fix $\widehat{X}_0 = X_0$. Along the same actual reports,

$$\mathrm{dist}_t(X_t, \widehat{X}_t) \le a_t\, \mathrm{dist}_{t-1}(X_{t-1}, \widehat{X}_{t-1}) + Cm_t^{-1/d}.$$

Consequently its squared terminal risk is at most

$$B_n + C \left[\sum_{t=1}^{n} \left(\prod_{j=t+1}^{n} a_j \right) m_t^{-1/d} \right]^2. \tag{23}$$

Orthogonality to the full-history audit residual justifies adding B_n. The approximate state is a codepoint named by the finite label, not an extra real register. This proof does not require that the distribution of the approximate state equal the acquired full-history law.

4.3. **Internal clocks and integer allocation.**

Lemma 4.1 (Positive-path clock separation)**.** *Let an autonomous finite-label machine use a reused Borel report row and independent update randomness, with deterministic halt status for each label. Suppose every physical conditional report law, at every finite history before halting, is equivalent to the same reference measure. If it halts exactly after n reports almost surely, its positive-probability label sets at phases $0, \dots, n$ are pairwise disjoint. Thus, writing m_t for their sizes,*

$$1 + \sum_{t=1}^{n} m_t \le M. \tag{24}$$

Proof. For a nonhalt label s, the edge to j is possible precisely when the set on which $k(j \mid s, y) > 0$ has positive reference measure. Equivalence makes this criterion independent of phase or of the entering physical conditional law. Every finite path consisting of possible edges therefore has positive conditional probability whenever its initial label is reached. To justify the product assertion, successively condition on each actual history: the next edge probability is a strictly positive measurable function there, and its integral over a positive-probability prefix is positive.

Suppose a nonhalt label were reached at phases $i < j < n$. From its occurrence at j, some finite label path of length $n - j$ reaches a halt label with positive probability, since the finite union of such paths has probability one. The same path from phase i would halt strictly before n, a

contradiction. Halt labels can occur only at phase n, and a label's halt status cannot change with phase. The initial support has at least one label. This proves (24). The proof uses common null sets, not equality of transition probabilities or a hidden phase counter. □

The lower in (11) follows by applying (8) and (20) at phase n: there are at most $M-n$ terminal labels. The class is empty below $n+1$, and a single label per phase gives feasibility at and above that threshold.

For the upper, let $B = M-n-1 \geq 0$ be the number of labels left after one per phase including the start. Choose $0 < z < 1$ and set

$$m_t = 1 + \lfloor B(1-z)z^{n-t} \rfloor, \qquad 1 \leq t \leq n. \tag{25}$$

Then $1 + \sum m_t \leq M$. For $B \geq 1$, $m_t \geq B(1-z)z^{n-t}$. Under (C), choose $z = \rho^{d/(d+2)}$ in (22); the geometric sum of $\rho^k z^{-2k/d}$ is finite and gives $r_n \leq CB^{-2/d}$. Under (P), choose $z = \rho^{d/(d+1)}$ in (23); the sum of $\rho^k z^{-k/d}$ is finite and gives the same order after squaring. For $B = 0$ the uniform diameter bounds suffice; for $B \geq 1$, B and $B+1 = M-n$ differ by at most a factor two. Use disjoint phase alphabets and their phase-specific rows to form one autonomous row. Every phase, codepoint and halt response is thereby counted. This finishes the proof of Theorem 2.2.

Remark 4.2 (What the theorem does and does not identify). Without the geometric product bounds, (22) and (23) retain their explicit continuation weights. They are upper bounds, not general matching weighted converses. The state-independent isometric examples in Companion G have nonsummable sensitivity and a genuinely different online curve despite unchanged acquired marginals. The present theorem does not erase that obstruction. Its horizon-uniform resource law is a sufficient stable class, not a universal classification of recursive compressibility.

5. A nonhomogeneous Gaussian filtering realization

Let the hidden state be $J_t \in \{1, \ldots, D_c\}$, where $D_c \geq 2$, and put $d = D_c - 1$. Let P be an invertible column-stochastic matrix with entries at least $\alpha > 0$. The raw call first reports $Y_t \mid J_{t-1} = i \sim N(m_i, \Sigma)$ and then draws J_t from column i of P. Take report dimension d, positive definite Σ, and affinely independent means $m_1, \ldots, m_{D_c}$. Larger report dimensions with a full-rank likelihood image work identically. The preparation has known law $x_0 \in K = P\Delta_{D_c-1}$. The terminal audit is the categorical vector $Z_n = e_{J_n}$, observed only after committing the answer.

The report-then-transition convention makes the prediction domain uniformly interior without modifying the actual Gaussian reports. If f_i are their densities, the raw positive instrument is

$$L_y = P\,\mathrm{diag}(f_1(y), \ldots, f_{D_c}(y)), \qquad \ell(y,x) = \sum_i x_i f_i(y), \qquad F(y,x) = P\frac{\mathrm{diag}(f(y))x}{\ell(y,x)}. \tag{26}$$

These are obtained from the hidden-state kernel. The coordinate tests are categorical probabilities, and unnormalized continuation tests are linear in them. Mixtures of prepared hidden states realize the simplex and its domain K. The coordinate audit separates the post-call predictions. Hence the declared predictive quotient and affine closure are realized, rather than postulated.

Theorem 5.1 (All positive invertible transitions)**.** *The raw experiment* (26) *satisfies Theorem 2.2 with acquisition dimension $d = D_c - 1$ and certificate (P). In particular its checkpoint and total-state autonomous excess risks are, uniformly for $n \geq 1$ and feasible M,*

$$\mathcal{R}_{\mathrm{cp}}(n,M) - B_n \asymp M^{-2/d}, \qquad \mathcal{R}_{\mathrm{aut}}(n,M) - B_n \asymp (M-n)^{-2/d}.$$

Constants may be chosen uniformly on compact parameter families having the stated positivity, invertibility and likelihood-rank bounds. No condition such as three times a total-variation mixing coefficient being less than one is required. The observation protocol is fixed; the infimum is over all observers of that protocol.

Proof. Every $x \in K$ has $x_i \geq \alpha$. On the simplex interior use Hilbert's metric

$$h(x,z) = \max_i \log(x_i/z_i) - \min_i \log(x_i/z_i).$$

On K it has finite diameter and is uniformly equivalent to the Euclidean tangent norm. Positive diagonal multiplication and subsequent normalization preserve h.

We record an elementary contraction for P. Interpolate by the normalized geometric path $x(s) \propto x^{1-s}z^s$, $0 \leq s \leq 1$. Differentiating $\log(Px(s))_i$ gives the average of $v_j = \log(z_j/x_j)$ with weights $w_{ij} = P_{ij}x_j(s)/(Px(s))_i$, minus a common normalization derivative. Since $(Px(s))_i \leq 1$, every row of weights dominates the same subprobability $\alpha x(s)$ of mass α. The oscillation of the row averages is thus at most $(1-\alpha)\operatorname{osc} v$. Integrating gives

$$h(Px, Pz) \leq (1-\alpha)h(x,z). \tag{27}$$

Initially x, z may be any strictly positive simplex points; approximation extends the relevant image statements to closed input faces. Combining diagonal isometry with (27) proves (P) for (26), with $a_t = 1 - \alpha < 1$ at every phase. The filter is genuinely nonlinear in its incoming belief.

It remains to derive actual acquisition mass. Write $w_i = x_i f_i(Y)/\ell(Y,x)$ and set $v_i = \log(w_i/w_{D_c})$ for $1 \leq i \leq d$. Then

$$v_i = \log(x_i/x_{D_c}) + (m_i - m_{D_c})^T \Sigma^{-1} Y - \tfrac{1}{2}(m_i^T \Sigma^{-1} m_i - m_{D_c}^T \Sigma^{-1} m_{D_c}).$$

Under the actual report law this vector is a mixture of D_c nonsingular Gaussian laws. Their covariance is fixed and positive definite by affine independence; their means range over a compact set because $x_i \geq \alpha$. Its density therefore satisfies $g_x(v) \leq C\exp(-c\|v\|^2 + C\|v\|)$ uniformly in x. The inverse log-odds map has Jacobian $\prod_{i=1}^{D_c} w_i$. Moreover

$$(\textstyle\prod_i w_i)^{-1} = (1 + \sum_{i=1}^d e^{v_i})^{D_c} e^{-\sum_{i=1}^d v_i} \leq Ce^{C\|v\|}.$$

Consequently the actual density of w on the entire simplex is uniformly bounded, including approaches to its boundary. The invertible affine transformation $w \mapsto Pw$ gives a uniform density bound on K, and hence (4). The dimension here is the rank of the acquired likelihood-coordinate map, not the report dimension or a formally added covariance.

The compact full-dimensional convex body K has Euclidean covers of size Ch^{-d}; metric equivalence gives the required Hilbert covers. Choose the lowest-index nearest codepoint to obtain a Borel rule. Finally $\|\ell(\cdot,x) - \ell(\cdot,z)\|_{L^1} \leq \|x - z\|_1$, proving report continuity. Every Gaussian density is positive everywhere, so all conditional report laws have the same null sets. The clock lemma applies. All estimates are uniform under the displayed compact parameter restrictions. □

This realization is an ordinary finite-memory nonlinear hidden-state filter with a one-step-ahead categorical forecast. For example the symmetric two-state transition with off-diagonal probability 0.01 has total-variation coefficient 0.98; a bound requiring $3 \cdot 0.98 < 1$ is unavailable, while (27) still gives a valid contraction. The resulting constant deteriorates as the transition approaches a reducible or deterministic limit; the theorem does not assert uniformity there. No report is discarded, accepted conditionally, or resampled without payment. The Gaussian change-of-variables argument is inherited in method from the earlier acquired-density theorem; the new scope here comes from the normalized causal defect and the same-report cone-metric implementation, not from claiming a new Gaussian density formula.

6. State-dependent noncommuting instruments

Let $D \geq 2$, $d = D^2 - 1$, and choose a real Hilbert–Schmidt orthonormal basis $F_1, \ldots, F_d$ of traceless Hermitian matrices. The prediction coordinate is $x_j = \operatorname{tr}(\rho F_j)$; its Euclidean distance is the Hilbert–Schmidt distance of density operators. This finite coordinate system will be verified by the raw instrument and audit. It is not memory available to the classical observer.

Let μ be normalized Lebesgue measure on the unit Hilbert–Schmidt ball of traceless Hermitian matrices H, and fix $0 < \eta < 1$, $0 < \lambda < 1$ and a known unitary U. Define

$$A_H = I + \eta H, \quad K_H = A_H^{1/2}, \quad T(\sigma) = \lambda U \sigma U^* + (1-\lambda)\,\mathrm{tr}(\sigma) I/D.$$

One raw call is the completely positive instrument

$$\mathcal{I}_H(\rho) = T(K_H \rho K_H)\,\mu(dH). \tag{28}$$

Since $\int H\, d\mu = 0$, it is normalized. Its report density and prediction update are

$$\ell(H,\rho) = 1 + \eta\,\mathrm{tr}(H\rho), \qquad F(H,\rho) = T\left(\frac{K_H \rho K_H}{\mathrm{tr}(A_H \rho)}\right).$$

The law depends on the entering state, and different effects A_H need not commute. Reports range over a Euclidean effect body, not a Haar group orbit. The physical channel and measurement together are one declared instrument call; any implementation that resolves its internal suboperations must charge their physical time separately.

Prepare a known $\rho_0 \in K = T(\mathcal{D}_D)$, where $\mathcal{D}_D$ is the density-operator body. Then every prediction has eigenvalues between $a = (1-\lambda)/D$ and $b = a + \lambda$. The bounded executable audit chooses I uniformly in $\{1, \dots, d\}$ after commitment, measures the two effects $(I_D \pm F_I)/2$, and returns $Z = dSe_I$. Positivity follows from $\|F_i\|_{\mathrm{op}} \le \|F_i\|_{\mathrm{HS}} = 1$. Directly,

$$\mathbb{E}[Se_I \mid \rho] = x/d, \qquad \mathbb{E}[Z \mid \rho] = x, \qquad \mathbb{E}\,\|Z\|^2 = d^2. \tag{29}$$

Only one basis test is performed, not a joint measurement of incompatible effects. All losses can be rescaled by a fixed dimension-dependent constant to lie in $[0,1]$ for a comparison theorem.

Lemma 6.1 (Positive congruence coordinates)**.** *For positive definite q, the map*

$$A \longmapsto \frac{A^{1/2} q A^{1/2}}{\mathrm{tr}(Aq)}$$

is a smooth bijection from positive definite matrices of trace D to positive definite density operators. Its inverse at σ is

$$B = q^{-1/2}(q^{1/2}\sigma q^{1/2})^{1/2} q^{-1/2}, \qquad A = DB^2/\,\mathrm{tr}(B^2). \tag{30}$$

On compact positive definite input domains all derivatives required for a change of variables are uniformly bounded, and the absolute Jacobian has a positive lower bound.

Proof. B is the unique positive definite solution of $BqB = \sigma$: conjugating by $q^{1/2}$ reduces uniqueness to the positive square root. Formula (30) normalizes its otherwise immaterial scale. Conversely any original A is recovered by that normalization. Positive square roots and inverses are smooth on the positive definite cone. Differentiating the inverse identities on the trace hyperplanes shows that the derivatives are inverse linear maps. Compactness supplies the uniform bounds and nonvanishing Jacobian. This elementary inverse is also used in the earlier positive-instrument realization; it is not claimed anew as a matrix identity. □

Theorem 6.2 (Weak effects followed by a positivity-improving channel)**.** *The raw instrument* (28)*, with the declared audit, verifies Theorem 2.2 with acquired dimension $d = D^2 - 1$ and certificate (P), for every $0 < \lambda, \eta < 1$. Constants are uniform on compact subintervals of those parameters and fixed finite D. The observer is classical and has no access to the physical system except the reports and the final post-commitment audit.*

Proof. The unnormalized instrument is linear in ρ, so all report/test continuations have the required finite affine closure. The audit coordinates separate density operators; their preparations provide a continuous realization. Positivity gives $1 - \eta \le \ell \le 1 + \eta$, a common report support

and $\|\ell(\cdot,\rho)-\ell(\cdot,\sigma)\|_{L^1}\le\eta\|\rho-\sigma\|_{\mathrm{HS}}$. Neither an eigenbasis chosen by the unknown state nor a preexisting information metric is assumed.

Use the cone metric $h(\rho,\sigma)=\log(M/m)$, where $m\sigma\preceq\rho\preceq M\sigma$ are the extremal order bounds. Positive congruence and normalization preserve this metric. It has bounded diameter $H_0=2\log(b/a)$ on K and is uniformly equivalent there to the Hilbert–Schmidt distance.

Here is a direct contraction estimate for T. For normalized positive ρ,σ, $m\le 1\le M$. Put $c=(1-\lambda)/(Db)\in(0,1)$. Since $T(\sigma)\preceq bI_D$,

$$[m+c(1-m)]T(\sigma)\preceq T(\rho)\preceq[M-c(M-1)]T(\sigma).$$

For example the lower follows by expanding $T(\rho)-mT(\sigma)$ and retaining its $(1-\lambda)(1-m)I_D/D$ part. The upper is the same argument applied to $M\sigma-\rho$. When $h(\rho,\sigma)\le H_0$, both m,M lie in $[e^{-H_0},e^{H_0}]$. The derivative of $\log((1-c)e^v+c)$ is bounded on this interval by

$$\vartheta=\frac{(1-c)e^{H_0}}{(1-c)e^{H_0}+c}<1.$$

Integrating between $\log m$ and $\log M$ proves $h(T\rho,T\sigma)\le\vartheta h(\rho,\sigma)$. Applied after the metric-isometric normalized congruence, this establishes (P). It does not require $3\lambda<1$.

For fixed $\rho\in K$, Lemma 6.1 applies to A_H on its compact effect ball, uniformly because $\rho\succeq aI_D$ and $A_H\succeq(1-\eta)I_D$. The linear parametrization by H has rank d. The final channel T is an invertible affine map on the trace-one hyperplane, with linear part λ times a unitary isometry. The posterior map therefore has a uniformly nonzero Jacobian. Its actual report density is bounded by $1+\eta$, so its image measure has bounded d-dimensional density. This proves the acquired small-ball estimate on every actual incoming row. The effect-ball boundary has zero reference mass, and no conditional renormalization is made. Global metric covers follow from Euclidean covers of the compact convex body K. All requirements of Theorem 2.2 have now been verified from (28). □

Corollary 6.3 (Noncontracting phases between bounded-gap improvements)**.** *In* (28)*, allow time-dependent known unitaries and channels that are either unitary ($\lambda_t=1$) or have $\lambda_t\in[\lambda_-,\lambda_+]\subset(0,1)$. Suppose every block of L consecutive phases contains an improving channel, and prepare an interior state with a fixed eigenvalue lower bound. With a common effect law and η bounded away from one, the same matching risk laws hold uniformly in the horizon. Individual normalized updates need not contract.*

Proof. A normalized effect changes a lower eigenvalue bound by at worst the factor $(1-\eta)/(1+\eta)$. An improving channel restores it to at least $(1-\lambda_+)/D$. The bounded-gap condition thus gives a common positive eigenvalue floor for all declared domains, including the initial incomplete block. Choose convex domains with these phasewise floors. Congruences and unitary phases are metric isometries; each improving phase contracts by a common constant less than one by the preceding proof. Products therefore satisfy (P). The inverse-map, report-continuity and cover constants stay uniform on the resulting compact interior domains, including unitary phases. This proves the assertion without replacing the block by an uncharged reset. □

7. Acquired strata with changing rank

The preceding quantum class stays in the positive definite interior. We now allow a physical rank change while retaining its actual probability. The tool is the curvature alternative, which does not require the same-report map to contract on a boundary stratum.

Proposition 7.1 (A weighted acquired-stratum profile)**.** *Suppose the affine certificate and (C) hold on a compact convex domain K, with b_t satisfying its product bound. Suppose each actual row*

law has the decomposition

$$\mu_{t,u} = \sum_{j=1}^{J} w_j \mu_{t,u}^{(j)}, \qquad w_j > 0, \quad \sum_j w_j = 1,$$

where, uniformly in t, u, the probability $\mu_{t,u}^{(j)}$ obeys a small-ball upper bound of exponent $d_j > 0$. Its support lies in a common compact set $E_{t,j} \subset K$ with at-most-m Euclidean covers of radius $C_j m^{-1/d_j}$. The component mark is visible in the report, or there is a common Borel assignment using the corresponding union codebook. Put

$$\Xi_w(m) = \sum_{j=1}^{J} w_j m^{-2/d_j}. \tag{31}$$

Then the checkpoint excess is comparable to $\Xi_w(M)$, and under the common-support autonomous interface its excess is comparable to $\Xi_w(M-n)$ for every $M \geq n+1$. Constants may depend on J, the displayed geometric and stability constants, but need not discard or normalize any w_j. If atoms are present their exact finite supports may instead be assigned a declared finite number of labels and zero residual profile thereafter.

Proof. For any global m-center set, its distortion on component j is at least $q_m(\mu^{(j)})$. Summing gives the lower $c\Xi_w(m)$, with no assumption that different components use disjoint centers. The actual full-history mixture has the same component weights and small-ball bounds, since these are uniform in its incoming prediction.

For $m \geq 2J$, allocate $\lfloor m/J \rfloor$ centers to each component; their union uses at most m centers and has distortion at most $C\Xi_w(m)$ under every row. For $m < 2J$, a single legal response has bounded error and $\Xi_w(m) \geq (2J)^{-a_*}$, where $a_* = \max_j 2/d_j$, so enlargement of a fixed constant gives the same upper. The representative after pooling lies in K, even if it lies on none of the acquired strata. Equation (21) applies to the union codebook and gives $r_t \leq b_t r_{t-1} + C\Xi_w(m_t)$.

The clock lower and row lower yield $c\Xi_w(M-n)$. In the allocation (25) choose $z = \rho^{1/(a_*+1)}$. Each geometric series $\sum_{k\geq 0} \rho^k z^{-2k/d_j}$ is bounded. The recursion thus gives $C\sum_j w_j (M-n)^{-2/d_j}$, including the minimal-budget case as before. Finite atoms can be covered exactly; the additional labels must be counted rather than deleted from the budget. This proves the stated extension. □

Here is a raw instrument with two genuinely different acquired ranks. Let $0 < \lambda < 1$ and first apply $q = T(\rho) = \lambda\rho + (1-\lambda)I_D/D$ to an arbitrary density operator, possibly singular. With probability $\omega \in (0,1)$ perform the weak effect instrument of Section 6 *after* this channel, with unnormalized output $K_H q K_H \, \mu(dH)$. With probability $1-\omega$ perform the rank-one instrument

$$q \longmapsto D \operatorname{tr}(P_v q) P_v \, \sigma(dv), \qquad P_v = vv^*, \tag{32}$$

where σ is the unitary-invariant probability on complex projective space. The branch mark is part of the report. The instrument is completely positive: its density is the map $q \mapsto DP_v q P_v$. Since $\int DP_v \, d\sigma = I_D$, both branches are normalized separately. The common preparation channel is charged within each instrument call; it is not a free regeneration or a discard-and-repeat protocol.

Theorem 7.2 (Repeated full-rank/pure-rank acquisition)**.** *Equip this raw instrument with the same separating audit as in* (29)*. If*

$$b_* := \omega\lambda^2 \frac{(1+2\eta)^2}{1-\eta} < 1, \tag{33}$$

then, for $n \geq 1$, the actual predictive law has its full-rank and rank-one components with masses exactly ω and $1-\omega$. With $d_f = D^2-1$ and $d_p = 2(D-1)$,

$$\mathcal{R}_{\mathrm{cp}}(n,M) = B_n + \Theta\big(\omega M^{-2/d_f} + (1-\omega)M^{-2/d_p}\big), \tag{34}$$

$$\mathcal{R}_{\mathrm{aut}}(n,M) = B_n + \Theta\big(\omega (M-n)^{-2/d_f} + (1-\omega)(M-n)^{-2/d_p}\big) \tag{35}$$

for feasible $M \geq n+1$. Constants have the dependencies in Proposition 7.1, including the distance of b_* from one. This is a total-risk statement with the actual baseline B_n, not the baseline of the nonsingular experiment.

Proof. Every q has eigenvalues at least $(1-\lambda)/D$, uniformly over the entire closed state body. In the weak branch Lemma 6.1 therefore supplies a bounded density of the full-rank output in d_f affine coordinates. Its support lies in a compact full-rank subset with a global d_f-dimensional cover. The pure branch has density $D\,\mathrm{tr}(P_v q) \leq D$ relative to σ. For a fixed pure projector P_w,

$$\|P_v - P_w\|_{\mathrm{HS}}^2 = 2(1-|v^*w|^2), \qquad \sigma\{\|P_v - P_w\|_{\mathrm{HS}} < h\} = (h^2/2)^{D-1}$$

for $0 \leq h \leq \sqrt{2}$. To verify the cap formula, normalize independent standard complex Gaussian coordinates. Their squared moduli are independent exponential variables; integrating their sum shows that the first normalized squared modulus has density $(D-1)(1-u)^{D-2}$ on $[0,1]$. For a ball centered off the projective orbit, choose any orbit point in it and enlarge the radius by two. This gives the required small-ball bound for arbitrary centers.

For the global pure-state cover, select an index with $|v_j| \geq D^{-1/2}$, choose its phase positive, and use the other $D-1$ complex coordinates as a finite chart. On a slightly enlarged such chart the projector map is Lipschitz and has $2(D-1)$ real coordinates. A finite collection of chart grids yields Ch^{-d_p} codepoints. No uniform simple spectrum or extraneous ambient exponent is used. The actual row measure is the indicated mixture with the original weights, because the branch coin is normalized before conditioning on its report. Full-history mixing over old predictions preserves those same weights and bounds.

For stability, use traceless tangent directions v. In the weak branch, with $u_H = K_H q K_H$ and $\ell_H = \mathrm{tr}(A_H q)$,

$$\|Du_H[v]\|_{\mathrm{HS}} \leq \lambda(1+\eta)\|v\|_{\mathrm{HS}}, \qquad |D\ell_H[v]| \leq \lambda\eta\|v\|_{\mathrm{HS}}, \qquad \ell_H \geq 1-\eta.$$

A normalized density operator has Hilbert–Schmidt norm at most one. Equation (17) therefore bounds half the weak-branch Hessian by $\lambda^2(1+2\eta)^2/(1-\eta)$ times $\|v\|_{\mathrm{HS}}^2$. Centering by I_D/D only subtracts an affine function from the potential and leaves its Hessian unchanged. In the pure branch the squared centered norm is the constant $1-1/D$, so its integrated potential is affine (in fact constant); its Hessian vanishes. The mixture has the bound (33). Derivatives extend to a neighborhood of the closed density body because q and the effects retain strict positive lower bounds.

The report density in either component is affine and uniformly positive on its component support; their mixture, with the visible mark and fixed ω, has common null sets at every phase. Total-variation continuity follows directly from the affine formulas. Proposition 7.1 now applies on the full convex density body. The true acquired rank changes between D and one, while the coarse observer's conditional means may have any rank; all of them are included in the curvature certificate. The physical baseline $B_n = \mathbb{E}\|Z_n - X_n\|^2$ is kept throughout. □

8. One task, complete resources, and attained error alternatives

8.1. **Task-compatible causal morphisms.** An experiment morphism consists of a parameter-independent causal simulator, a legal strategy lift, report and task-readout transformations, a clock map, and a declared finite internal state. Each virtual report must be emitted before the target decision using it. Raw acquisitions and physical time are mapped as cumulative nonnegative costs.

A report transformation without a strategy lift or without a compatible physical audit is not such a morphism. The simulator may not use the later environment-owned audit to generate earlier reports.

Proposition 8.1 (Composition in a common bounded task)**.** *Suppose a target observer uses at most M persistent states and a task-compatible simulator at most S internal states. Suppose the complete joint laws of the stopped transcript, the observer's fresh randomization, its committed answer and the protected scored audit differ by total variation at most ε. The composed observer uses at most MS states and, for loss in $[0, L]$, its risk differs by at most $L\varepsilon$. For a chain of simulators, errors add (capped at one), state counts multiply, and the declared call/clock maps compose. A reverse risk inequality requires a reverse task-compatible simulator.*

Proof. The runtime state is the pair of observer and simulator labels. The legal strategy lift and report deadlines make its transitions causal. The conditional distribution of each fresh randomization can be applied to a common coupling without increasing total variation. Thus a coupling fails for the complete scored joint law with probability at most ε, and the bounded loss difference is at most $L\varepsilon$. The union bound and induction give the chain statement. No inversion of a one-way simulation is used. A statistic in $[-1, 1]$ instead has expectation discrepancy bounded by 2ε under the convention $\mathrm{TV}(P, Q) = \sup_E |P(E) - Q(E)|$. □

Persistent bits cost $\lceil \log_2(MS) \rceil$, not the simulator's program description or its temporary numerical workspace. Internal counters belong to S or M, never to an uncharged clock map. This finite-horizon proposition is not a stopping-time, infinite-horizon, pressure or large-deviation equivalence without further hypotheses.

8.2. **Certified finite-precision execution.**

Proposition 8.2 (Precision with a boundary certificate)**.** *Fix a finite ideal program from the proof of Theorem 2.2. For every phase t and every positive-probability incoming label, suppose the ideal actual pre-quantization row law assigns mass at most $\omega_t(h)$ to the h-neighborhood of its cell boundaries. Suppose a digital implementation, while its old label agrees with the ideal one, computes that cell decision with coordinate error at most h_t outside an event of unconditional probability at most ξ_t. Suppose it approximates each final readout within v. If the primitive kernels or calibration law are additionally coupled with the ideal complete experiment with error ζ, then*

$$|R_{\mathrm{digital}} - R_{\mathrm{ideal}}| \le L \min\{1, \zeta + \sum_t [\omega_t(h_t) + \xi_t]\} + Cv \tag{36}$$

for bounded squared tasks and a fixed bounded response set. No extra persistent states are required if the compiled label set is unchanged. Numerical workspace, report precision and stored constants must nevertheless be paid separately.

Proof. Use the same reports and fresh randomness until the first label mismatch. Outside the primitive coupling failure and numerical exceptional events, that first mismatch can occur only when the ideal pre-quantization value lies within h_t of a boundary. The unconditional mass of that event is bounded by the row certificate after summing over old labels. Restricting to histories before the first mismatch only decreases this unconditional probability; no density bound conditioned on that restriction is assumed. The union bound gives the displayed minimum. If labels never disagree, the bounded squared loss is Lipschitz in the final response and changes by at most Cv. This proves (36) without claiming TV proximity between a continuous report and its quantized numerical representation. □

For a Euclidean grid in a bounded d-dimensional coordinate body and actual row densities bounded by A, a mesh with $O(m^{1/d})$ hyperplanes has $\omega_t(h) \le CAm^{1/d}h$, capped at one. A Voronoi implementation can instead bound each of its finitely many bisector slabs, giving the conservative

certificate CAm^2h. Boundary tie rules are fixed Borel rules. In singular components one must supply the corresponding component-wise boundary certificate or choose chart grids; ambient Lebesgue density is not available for free.

The raw Gaussian and positive-effect models with computable, quantitatively separated parameters admit explicit certified arithmetic. For Gaussian reports, truncate computation outside a radius R only as a *charged failure event*: a Gaussian tail bound on the known finite mixture makes its total probability at most a chosen ξ, uniformly over the incoming labels. For example $R = C(1 + \sqrt{\log(n/\xi)})$ suffices with parameter-dependent constants. The likelihood ratios are evaluated in log coordinates on that bounded event. Positive spectral floors in the quantum model give certified inverses and square roots by polynomial or interval iteration. Rational codepoints can be projected into the declared positive compact domain with a certified tolerance. These operations approximate the chosen program, not an unobservable true parameter.

A deliberately conservative accounting for codebook implementations with k coordinates, M total labels and p-bit constants is: $O(kMp)$ read-only codepoint bits, the separately specified primitive parameter description, a transition/phase description of $O(M \log M)$ bits, and a final response table included in the same $O(kMp)$ term. Naive nearest-center evaluation uses $O(Mk)$ arithmetic operations per report in addition to primitive evaluation; $O(k^2p + \log M)$ working bits suffice for serial distance and matrix calculations at a declared internal accuracy, with any larger interval-iteration workspace stated explicitly. Counts refer to shared read-only constants rather than one copy per traversed edge. The report-precision oracle and its time are part of the implementation interface. Calibrated-cell construction under (C) additionally requires certified forward integrals for its representatives; mere Borel existence does not supply those computations. No matched program-length, workspace or arithmetic-time lower is inferred from the label theorem.

8.3. **A matching calibration–deficiency law with a counted retained bit.** A positive upper error budget by itself cannot supply a minimax lower. We next give an attained physical alternative and count the memory needed to retain its useful report. Fix any core experiment of Theorem 2.2 in dimension d, with $n \geq 1$ core report calls. Independently prepare a fair bit $U \in \{-1, 1\}$ and an unknown fixed calibration offset $\theta \in \{-\delta, \delta\}$. The initial paid action BIT reports U with probability $1 - a$ and the erasure symbol with probability a, where $0 \leq a \leq 1/2$. It must be followed by exactly n passive core FILTER calls. The answer is committed, and then one joint audit returns (Z_n, U, θ). The loss is the sum of the three squared coordinate losses. The true sign of θ is unavailable to the program. The preparation, bit call, core calls and final audit are

$$N = n + 3 \tag{37}$$

physical calls. The initial action and the later FILTER actions must be selected by the machine's labels; no action-availability signal supplies a phase. Programs issuing a different prescribed action word are not legal for this experiment.

Compare this experiment to the target whose initial bit report is exact, with the same protected audit and core experiment. The simulator must emit its virtual bit before any core reports. Put $\varepsilon = a/2$.

Theorem 8.3 (Attained joint law and full memory charge)**.** *The uniform causal deficiency of the erased-bit interface to the exact-bit target, with its audit protected and its report deadline fixed, is exactly ε. For any fixed bounded range of δ, the joint minimax checkpoint risk satisfies*

$$\mathcal{R}^{\mathrm{joint}}_{\mathrm{cp}}(n, M, \delta, \varepsilon) - B_n \asymp M^{-2/d} + \delta^2 + \varepsilon, \qquad M \geq 1. \tag{38}$$

The legal autonomous class is nonempty exactly when $M \geq n + 2 = N - 1$. On that class, set

$$Q(n, M) = \max\{1, M - 2n - 1\}.$$

Then

$$\mathcal{R}^{\mathrm{joint}}_{\mathrm{aut}}(n, M, \delta, \varepsilon) - B_n \asymp Q(n, M)^{-2/d} + \delta^2 + \varepsilon. \tag{39}$$

Constants depend only on the core certificate, the task bound and the fixed offset range; a itself is retained in the risk. Neither the true bit on erasure nor the calibration sign is passed to the observer. In terms of physical acquisitions, substitute $n = N - 3$ *in* (39).

Proof. On a nonerasure, transmit the bit; on erasure, emit a fresh fair bit. This stateless simulator has disagreement probability $a/2$ with the physical audit bit, and otherwise matches the target joint law, including all later core observations. Its total variation error is $a/2$. Conversely, conditional on erasure, every admissible simulator's pre-core information is independent of U, even with arbitrary additional classical memory. It guesses incorrectly with probability $1/2$. The event that its virtual bit disagrees with the later protected physical bit has probability at least $a/2$ and probability zero under the exact target. This proves the exact deficiency. Allowing the simulator to replace that audit would be a different comparison.

For either observer class, averaging the squared offset loss over the two signs gives at least δ^2; the response zero attains this. Even the complete available history has bit error a on erasure and zero otherwise. The independent core's all-label last-row converse remains valid with the bit and observer randomizations included in the old label. It yields $cM^{-2/d}$ at a checkpoint. Summing the three lower bounds under the common uniform sign prior, and then using minimax domination, proves the lower in (38). For the upper, impute an independent fair bit on erasure and keep the resulting sign b. The fixed response $(1-a)b$ has bit risk $2a - a^2 \le 2a$. Use two copies of a core checkpoint code with at most $\lfloor M/2 \rfloor$ labels. For $M \ge 2$ this gives the upper, and for $M = 1$ a constant response gives the same order after enlarging constants.

For an autonomous machine, the initial Bit label cannot recur at a Filter phase because its action row is fixed. All subsequent core-phase report laws have common support, so Lemma 4.1 separates the positive-probability label sets at the phases after the bit and after each core call, including halt. Thus $n + 2$ labels are necessary and one label per phase suffices for legality.

There are now two possibilities. If any of these post-bit phase sets, including halt, contains only one positive-probability label, all retained information about U has been lost at that phase. The physical core and its future reports are independent of U, and there is no external decoder transcript or persistent seed. The bit error at the final response is therefore at least one. If every nonterminal post-bit phase contains at least two labels, they use at least $2n$ labels, besides the initial label. Hence the final phase has at most $M - 2n - 1$ labels, and the core last-row converse yields $cQ(n, M)^{-2/d}$. If its final phase instead has one label, the first case applies. In either case the joint risk above B_n is bounded below by a constant times $Q^{-2/d} + a$, plus δ^2: in the first case $Q^{-2/d} \le 1$ and $a \le 1/2$; in the second use the core lower and the complete-history bit lower. This also covers randomized encoders, since all phase sets and independence statements are under their actual laws.

For the upper when $M \le 2n + 2$, ignore the bit and use a legal constant-output core program, giving bounded excess; here $Q = 1$. For $M \ge 2n + 3$, reserve one initial label, two imputed-bit labels after the first call, and two copies of every later core-phase label. Choose widths k_t with

$$1 + 2 + 2\sum_{t=1}^{n} k_t \le M.$$

The core construction may therefore use total budget $M_0 = 1 + \lfloor (M-3)/2 \rfloor \ge n + 1$. Its excess is at most $C(M_0 - n)^{-2/d}$, and $M_0 - n$ is comparable to Q by fixed constants. Keep the sign tag throughout and output $(1-a)b$, and output zero for the offset. The resulting excess is at most $CQ^{-2/d} + 2a + \delta^2$. Every tag, phase and halt label has been charged. This proves (39). □

The earlier articles already use the elementary erasure and hidden-offset devices. The new assertion here is the two-sided retained-bit clock threshold and its combination with the broader Bayesian filter class. Formula (39) is an attained same-task law, not a universal lower bound for every calibration or simulator error of those sizes. The exact count N is a prescribed exposure protocol for a changing physical state, not an unrestricted at-most-N static experimental-design problem.

9. Comparison and mathematical scope

9.1. **Neighboring theories.** Static quadratic quantization supplies the optimal finite-center problem and its mass/cover bounds [1]. Conditional projection is standard probability, and the convex perspective used in Section 3 is a standard convexity-preserving operation [2]. The broader relation between convex information, scoring rules and Bregman defects is classical [3]. None of these ingredients, by itself, identifies the cost of reusing an outgoing label across distinct incoming posterior rows.

Borkar, Mitter and Tatikonda [4] formulate sequential vector quantization of a stationary Markov source as a partially observed stochastic control problem and characterize optimal encoding schemes by dynamic programming. Foster, Gray and Dunham [5] use a finite-state quantizer whose old state selects a codebook and whose current codeword helps determine the next state. These are direct predecessors of the finite-state viewpoint. The present observer receives a noisy instrument report rather than the current source or exact posterior. Theorem 2.2 gives a lower against every legal hard-state machine and a matching implementation with its internal deterministic deadline counted; it is not a claim that finite-state quantization or its control formulation is new.

Recursive marginal quantization [6] studies accumulated discretization error in recursively quantized Markov dynamics, including the Euler scheme of a diffusion. Its one-step error propagation is close in spirit to (23). Here the exact normalized defect identifies the missing old information, and the separate pooling term quantifies compatibility across predecessor labels. The acquired-mass converse and complete clock charge are additional assertions; the recursive upper alone is not presented as a new general quantization principle.

Nonanticipative and sequential rate-distortion theories [7, 8, 9] treat causal reproduction laws, information-rate lower bounds and, in Gaussian settings, filtering realizations and finite-horizon allocation. Their charged object is information rate, not a fixed total alphabet containing control phases and halt labels. In particular an information-rate allocation is not substituted for Lemma 4.1. Conversely the present finite-state theorem does not provide a coding theorem or the semidefinite characterization of Gaussian sequential rate-distortion.

Atar and Zeitouni [10] use projective contraction in nonlinear-filter stability. The metric mechanism in the raw realizations is part of that mature tradition; its elementary estimates are included to fix the constants and observation timing. Kara and Yüksel [11] establish near-optimal finite-memory feedback policies through filter-stability methods in partially observed control. Ghomi, Linder and Yüksel [12] obtain stationary optimal zero-delay codes and finite-window approximation for linear vector Markov sources under an infinite-horizon average criterion. The present results instead fix the physical report protocol, charge an exact internal deadline, and derive the hard-label exponent from the actual conditional report image. They do not replace those control or infinite-horizon conclusions.

Theory	Observed object	Charged memory/information	Main kind of conclusion
Sequential vector coding [4]	Markov source at encoder	Causal code/control structure	Optimal encoding characterization
Finite-state vector coding [5]	Current source vector and code state	State-selected codebooks	Design and storage/performance analysis
Recursive quantization [6]	Recursively approximated dynamics	Marginal codebooks	Accumulated approximation error
Nonanticipative RDF [8, 9]	Causal source/reproduction laws	Information rate	Distortion bounds and filtering design
This theorem	Noisy positive-instrument report and old label	All persistent phases, labels and halt states	Exact defect, all-observer converse and matching order

The table compares information patterns and assertions, not journal rank or historical priority. The published versions are cited where verified; absence of an exact bibliographic match is not a priority certificate.

9.2. **Relation to the retained foundations results.** The complete earlier articles are separate companions, not prerequisites to reading the present proof. The earlier posterior-pooling theorem already made conditional barycenters and Jensen losses operational for controlled noisy experiments. The earlier continuation-curvature theorem already obtained a charged-clock $(M-n)^{-2/d}$ law under a Wasserstein continuation certificate, including a stronger history-dependent action comparison than is claimed here. The Gaussian log-odds density estimate, positive congruence inverse, elementary erased-bit/offset devices, and integer phase allocation are inherited tools and have been explicitly re-proved or credited where used.

The present extension does not claim a new exponent merely because the same expression occurs again. It supplies the likelihood-normalized one-step perspective and its exact row/pooling converse, then obtains matching laws through a different same-report posterior-transport certificate. That certificate covers every positive invertible finite-state transition and every positive amount of the stated quantum improvement, including parameters outside the older sufficient 3κ block criterion. The rank-changing mixture combines that identity with a weighted acquired-stratum profile. The retained-bit theorem determines the additional autonomous phase bottleneck in the same scored experiment. The earlier relative-Haar theorem remains independently useful: it treats isometric updates for which summable stability fails and gives a different exact online law.

The General Theta problem is the connection between executable tests, causal implementation, acquired geometry and resources. Theorem 2.2 proves that connection for a specified finite affine test class with informative likelihoods. It is neither a classification of all physical experiments nor a replacement of foundations by a single realization. The two raw models are independent verifications of the same general theorem; they are not premises in its proof.

Two substantial mathematical targets remain. First, one needs matching converses and implementable recursions for nonsummable, state-dependent continuation sensitivity, without inserting a global compatible cover as an assumption. Second, one needs a joint learning-and-memory law for unknown kernels under an explicitly fixed action and calibration interface. Unbounded stopping, arbitrary nondominated supports and fully optimal computation resources also require separate proofs. The finite-horizon results here do not acquire those conclusions by notation or by a limiting argument without additional uniform estimates.

Appendix A. Operational refinements of the retained transport results

This appendix records precise supplements to the complete relative-report article in Companion G. Its exact circle and projective-orbit arguments are not shortened or replaced. The refinements do not extend a state-independent theorem to informative reports; that extension is proved separately in the main text.

A.1. **Scheduled and autonomous information patterns.**

	Scheduled observer	Autonomous observer
At update	Old label, current report, fresh randomness and phase	Old label, current report and fresh randomness only
Rows	A prescribed row at each phase	One reused row; phase must be encoded in labels
Budget	$\lvert S_t\rvert \le m_t$ at each phase	Initial, intermediate and halt labels all count in M
Deadline	Externally scheduled	Must halt exactly internally; reports remain available
Relative circle	Excess $1-\prod_t \beta_{m_t}$	$1+\sum_t m_t \le M$, balanced widths attain optimum
Stable Bayesian class	Bounds (22) or (23)	Excess comparable to $(M-n)^{-2/d}$

On the circle, $\beta_m = (\sin(\pi/m)/(\pi/m))^2$ with $\beta_1 = 0$. For $M-1 = nq+r$, $0 \le r < n$, the retained autonomous excess is $1-\beta_q^{n-r}\beta_{q+1}^r$, with feasibility $M \ge n+1$. Exact real comparisons to regular arc boundaries are Borel operations. A bit program must use the separately charged report-precision oracle and approximate constants; it is not an exact implementation of arbitrary real arithmetic.

The static fractional-cell rearrangement used there follows without an assumption of deterministic cells. For a measurable membership $0 \le f \le 1$ of circle mass p, rotate the first moment to the real axis. Among such memberships, the integral of $f(\theta)\cos\theta$ is maximized by taking the largest values of $\cos\theta$, namely the centered arc of mass p. Subtract its indicator: the difference has zero integral and opposite sign to $\cos\theta - \cos(\pi p)$, proving the inequality directly. The magnitude of the normalized first moment is at most $\sin(\pi p)/\pi$. This is the rearrangement step in the retained static proof, consistent with the circle quantization literature [13].

Its exact local saturation statement requires an attaining direction, an attaining row quantizer, and identical centroids at every common outgoing label. Static compact quantization attains its center infimum, but a directional coefficient defined over a larger family may only be an infimum. Exact circle arcs supply global compatible recursion; the higher-dimensional orbit cover supplies only a matching rate. A zero-energy row contributes no constraint, and a later zero product factor does not force earlier equalities. Repeated or unused codepoints give at most the declared number of labels. The response set is the closed unit ball, or the appropriate convex hull, so the scaled terminal readout is legal.

A.2. **A modular spectral projection estimate.**

Lemma A.1 (Uniform gap and projector control)**.** *Let V be a traceless Hermitian $D \times D$ matrix with $\|V\|_{\mathrm{HS}} = 1$, and write its eigenvalues decreasingly. There is an index $j(V) \in \{1, \dots, D-1\}$ with*

$$\lambda_{j(V)} - \lambda_{j(V)+1} \ge g_D := \frac{1}{\sqrt{D}(D-1)}.$$

For any two conjugates W, W' of V, and their top-$j(V)$ projections P, P', one has

$$\|P - P'\|_{\mathrm{HS}} \le g_D^{-1}\|W - W'\|_{\mathrm{HS}}. \tag{40}$$

Consequently there exist $C_D, r_D > 0$ such that for every such V, every Hermitian center Z, and $0 < r \le r_D$,

$$h\{U : \|UVU^* - Z\|_{\mathrm{HS}} \le r\} \le C_D r^{2(D-1)},$$

where h is Haar probability. The constants are uniform over repeated eigenvalues.

Proof. One eigenvalue has absolute value at least $D^{-1/2}$ and trace zero places another on the other side of zero. Their range is at least $D^{-1/2}$, so an adjacent gap is at least g_D. Choose the smallest index meeting this bound to define $j(V)$ measurably. The top projection is unique despite possible repeated eigenvalues within either separated cluster.

The spectral gap implies

$$\mathrm{tr}(W(P - P')) \ge g_D(j - \mathrm{tr}(PP')), \quad \mathrm{tr}(W'(P' - P)) \ge g_D(j - \mathrm{tr}(PP')).$$

For the first inequality, expand in an eigenbasis of W; the trace moved out of its top subspace equals the trace moved in below it, and each such unit loses at least g_D. Adding the two inequalities and using $\|P - P'\|_{\mathrm{HS}}^2 = 2(j - \mathrm{tr}(PP'))$ proves (40) by Cauchy–Schwarz, including its zero case.

If the radius-r ball meets the orbit, fix one conjugate in it. All projections of other conjugates in the ball lie within $2r/g_D$ of its top projection. Haar top projections are uniform on the complex Grassmannian of rank j. In its graph chart $A \in \mathbb{C}^{(D-j)\times j}$, normalizing a complex Gaussian frame and writing its lower block as AB gives chart density proportional to $\det(I + A^*A)^{-D}$. To verify the exponent, the substitution in the lower block has real Jacobian $|\det B|^{2(D-j)}$; integrating over the upper block after the change $(I + A^*A)^{1/2}B$ contributes determinant power $-D$. Near the origin the density is bounded and projector distance is comparable to $\|A\|_{\mathrm{HS}}$. A radius-s neighborhood therefore has mass at most $C_{D,j}s^{2j(D-j)}$. Unitary invariance centers the same estimate anywhere. Since $2j(D-j) \ge 2(D-1)$, and there are finitely many j, take the maximum of the constants and the minimum radius. This proves the uniform claim without assuming a simple spectrum. □

For a rank-one projector, a Borel section modulo phase is explicit: choose the first index j with $P_{jj} \ge 1/D$ and set $v = Pe_j/\sqrt{P_{jj}}$. Then v_j is real positive and $vv^* = P$. A finite-chart Gram–Schmidt rule, choosing the first nonzero residual at each step, completes v to a unitary; adjusting the last unused column phase gives determinant one when needed. The charts cover the orbit and their tie sets are Borel. The projective dimension in Companion G is $2(D-1)$ for the declared linear audit, not $D^2 - 1$ by fiat. Its observer is classical and exact higher-dimensional product saturation is not asserted.

For the contrast normalization used there, $\rho = I_D/D + rc_D X$ with $c_D = \sqrt{1 - 1/D}$ and $k_q = D^2 - 1$. Selecting one Hilbert–Schmidt basis effect after commitment gives

$$\mathbb{E}[Se_I \mid \rho] = (rc_D/k_q)x.$$

The basis operator norm need only be at most one. Constants depending on D are distinct from those depending on r (or other physical contrast); the scored audit bound deteriorates as contrast tends to zero. Sphere coordinate dimension is denoted $k_s = d_s + 1$, not the same symbol as this quantum audit dimension.

A.3. **Singular mass, temporal access, and digital conventions.** For retained hidden Haar reports the actual predictive law is $w_n\mu + (1 - w_n)\delta_0$. Its full-history baseline is $B_n = V - w_n$; total risk is that baseline plus the surviving-mass compression loss, not the un-erased baseline $V - 1$ plus a renormalized law. Inactive labels and their phase copies are counted. Lemma 4.1 supplies the common positive-path-support version of the deadline argument when phase report probabilities differ but their null sets agree. A visible suppression pattern with disjoint supports can reveal a clock and is outside that lower bound.

The retained fixed-start access theorem can be sharpened for one fixed model x by applying its family theorem to $\{x\}$; the remainder is then the singleton remainder $B_N(\{x\})$, with its own

constants, not the original family remainder silently relabeled. The initialization calls, Boolean gate and prefix clock are extra resources of that tester and remain outside the original observer budget. An ungated scalar discrepancy can still cancel even when all rows are accessible. The corrected physical crossing convention is $C_N \le N$ and the older shifted index is $K_N = C_N + 1 \le N + 1$; the predictable starts are $T_{j-1} < N$.

For finite-bit circle execution the mismatch bound is $\min\{1, C(M-1)h\}$, not its uncapped linear expression. Sine, cosine and response constants can be shared read-only data; copying them into every row is a different program count. Complete-law simulator comparison includes the target's fresh randomization. Protected-audit lower bounds based on a $[-1, 1]$ statistic use the factor two under the stated TV convention. These are finite-horizon results without a regenerative decomposition, not assertions that a physical process has no preparation. The corrected temporal companion is distinguished from its originally reviewed source; both original source and report remain preserved independently.

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